

Project Design Phase

Solution Architecture

Date	03 October 2026
Team ID	6ab7f10630f9767f87728ca3
Project Name	Import Data Using Transform Maps (Spreadsheet)
Team Leader	E Dilli Kumar
Team Members	R Vinish, K Jerina, B Mohanavalli, P Sri Ragavan
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between business problems and technology solutions. For this project, the business problem is the slow and error-prone manual entry of spreadsheet data, and the architecture goals are to:

- Find the best technical solution: import spreadsheet data automatically using Import Sets and Transform Maps.
- Describe the structure, characteristics and behavior of the solution to project stakeholders.
- Define the features, development phases and solution requirements (upload, mapping, transform, logging, reporting).
- Provide specifications according to which the solution is defined, managed and delivered.

Solution Architecture Diagram:

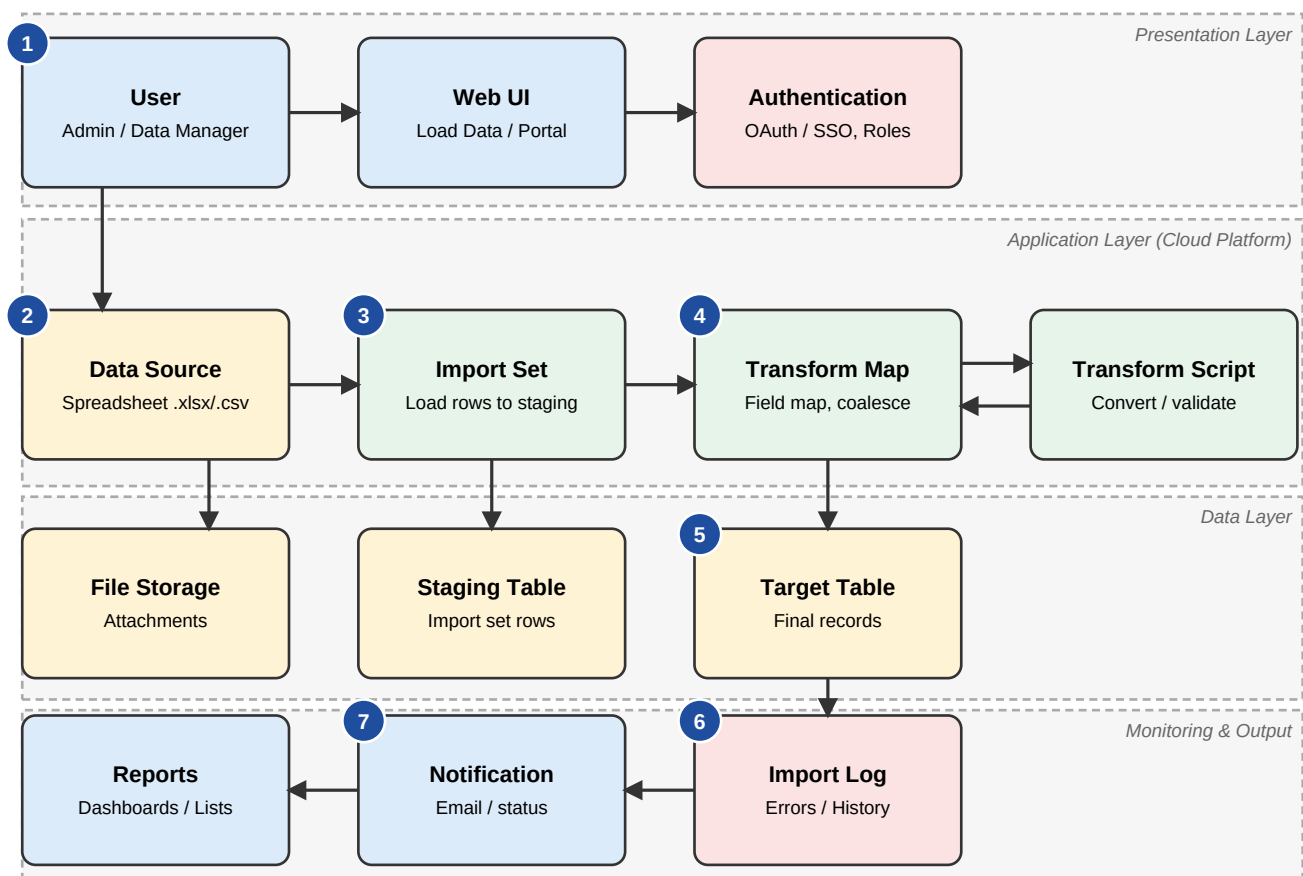


Figure 1: Architecture and data flow of Import Data Using Transform Maps (Spreadsheet)

Data Flow Description:

Step	Component	Description
1	User	The admin or data manager logs in (authenticated with roles) and opens the Load Data page in the web UI.
2	Data Source	The spreadsheet (.xlsx / .csv) is uploaded and stored as an attachment; file type, size and headers are validated.
3	Import Set	The rows of the spreadsheet are loaded into a staging (import set) table without changing live data.
4	Transform Map	Source columns are mapped to target fields; coalesce on key fields finds existing records; transform scripts convert and validate values.
5	Target Table	New records are inserted and existing records are updated in the target table.
6	Import Log	The result of each row (inserted, updated, skipped, failed) and the failure reason are recorded in the import log and transform history.
7	Notification & Reports	The user receives the import status and can view the imported data, summary reports and error reports on dashboards and lists.

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>